Abdelrahman Rizk

AI & Machine Learning Engineer

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Professional Summary

AI & Machine Learning Engineer experienced in building production-grade RAG architectures, agentic LLM workflows, and real-time Computer Vision solutions. Proficient in Python, SQL, C++, PyTorch, TensorFlow, Scikit-Learn, and ensemble modeling (XGBoost, Random Forest). Experienced in engineering vector retrieval pipelines (LangChain, ChromaDB, Ollama) and real-time object detection systems (YOLOv8, OpenCV, Supervision). Skilled in containerized deployments using Docker, FastAPI, and Streamlit on Linux (Fedora).

Experience

AI AUTOMATION — **Cairo, Egypt**

Digital Egypt Pioneers Initiative (DEPI) – MCIT (200+ Intensive Training Hours) — *Jul 2026 – Present*

- Engineered automated AI agent workflows integrating LangChain, vector retrieval (RAG), and custom tools to automate data extraction and report generation.
- Constructed end-to-end data pipelines connecting local LLMs (Ollama) with vector indexes in ChromaDB, achieving sub-second local retrieval response times.
- Collaborated in agile development sprints, maintaining version control workflows and deploying containerized prototypes on Linux.

Technical Projects

Autonomous AI Research Assistant (RAG) | *Python, LangChain, ChromaDB, Ollama, Streamlit* *2026*

- Built a local RAG engine parsing scientific PDFs, chunking technical prose, and generating dense vector embeddings stored in ChromaDB.
- Eliminated third-party API dependencies and data leaks by serving quantized open-weights models through Ollama on local hardware.
- Implemented strict source attribution with page-level citations in Streamlit, cutting factual hallucination rates significantly.

Egyptian Real Estate Valuation Platform | *Python, XGBoost, Scikit-Learn, Streamlit* *2026*

- Developed a production pricing engine employing a stacking ensemble of XGBoost, Random Forest, and Ridge Regression models.
- Engineered domain-specific geospatial features and imputation pipelines for missing property specs, minimizing Mean Absolute Error.
- Deployed an interactive Streamlit UI offering real-time price estimations and SHAP-based feature importance explainability.

Real-Time PPE Safety Compliance Detection | *YOLOv8, OpenCV, Supervision, Python* *2026*

- Trained and tuned a custom YOLOv8 model detecting hardhats, safety vests, and personnel across high-resolution surveillance video.
- Combined multi-object tracking via Supervision with custom spatial association logic to trigger instantaneous non-compliance alerts.

HR Analytics & Attrition Predictor | *Python, Scikit-Learn, Docker, Streamlit* *2026*

- Trained a cost-sensitive classification pipeline handling severe class imbalance via SMOTE to maximize minority-class recall and F1 score.
- Containerized the prediction microservice using Docker with optimized image layers for cross-platform deployment.

Education

Helwan University — **Cairo, Egypt**

Bachelor of Computer Science — *Expected Graduation: 2028*

Certifications & Programs

Linear Algebra for Machine Learning and Data Science *Oct 2026*

- Vendor / Provider: DeepLearning.AI & Coursera — *Foundations of ML Mathematics*

AI Automation (200 Hours) *Jul 2026 – Present*

- Vendor / Provider: Digital Egypt Pioneers Initiative (DEPI) – Ministry of Communications and Information Technology (MCIT)

AI Diploma – EraaSoft Academy — Aug 11, 2026

Machine Learning (25-Hour Offline Program) *Jan 2025 – Feb 2025*

- Vendor / Provider: Creativa Innovation Hubs, Pixels Egypt & ITIDA / MCIT — *Project: Spaceship Titanic*